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PRINTING APPARATUS FOR PRINTING DIRECTLY ONTO CONTAINERS

Abstract

An apparatus includes a plurality of print heads configured to print directly on a curved surface of each of the plurality of containers and a plurality of container holders for retraining each of the plurality of containers. Each of the plurality of container holders is configured to rotate a container retained therewithin and to move the container linearly relative to the plurality print heads in a continuous motion to maintain a print location on the curved surface of each of the plurality of containers at a substantially constant perpendicular distance from each of the plurality of print heads during a printing process. The apparatus may also provide for movement of the print heads during the printing process.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/690,153, filed Mar. 9, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/839,181, filed Apr. 3, 2020, the contents of all of which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to printing apparatuses, and more particularly, to printing apparatuses configured to print directly onto a container.

BACKGROUND

[0003] Currently, there are printing apparatuses for printing onto containers in the marketplace. However, these printing apparatuses are configured to perform conventional printing on mostly flat labels or flat faces of containers. Printing directly onto curved surfaces of a container is difficult in as much as the distance from the print head and the surfaces change over a curved surface. For example, conventional print heads only dispense the ink a few millimeters if they are vertically positioned. Thus, the print head needs to be as close as possible to the surface of the container to retain print quality, e.g., the clearness of images or texts. If a distance greater than a few millimeters exists between the print head and the printing surfaces, which is often the case along a curved surface, the printing process may be difficult, and the print quality may not be satisfactory. [0004] Accordingly, there exists a need for printing apparatuses to address effective printing on curved surfaces such as rounder curved surfaces.

SUMMARY

[0005] According to an embodiment of the present invention, a method and apparatus is provided for printing on a curved container surface. The apparatus includes a plurality of print heads configured to print directly on a curved surface of each of the plurality of containers and a plurality of container holders for retraining each of the plurality of containers. Each of the plurality of container holders is configured to rotate a container retained therewithin and to move the container linearly relative to the plurality print heads in a continuous motion to maintain a print location on the curved surface of each of the plurality of containers at a substantially constant perpendicular distance from each of the plurality of print heads during a printing process.

[0006] Additionally, in another embodiment, the print head may move to allow for printing on a round container.

[0007] These and other aspects of the present invention will be better understood in view of the drawings and following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a top view of a printing apparatus, according to an embodiment of the present invention.

[0009] FIG. 2 is a side view of a container holder of the printing apparatus in FIG. 1.

[0010] FIG. 3A-3E are top views of a container and a print head of the printing apparatus in FIG. 1, as the container continuously moves and rotates during a printing process.

[0011] FIG. 4 is a top view of a printing apparatus in accordance with a further embodiment of the

present invention.

[0012] FIG. 5 is a side view of the container holder of the printing apparatus of FIG. 4.

[0013] FIG. 6 is a schematic depiction of movement of the container about the turret during the printing process.

DETAILED DESCRIPTION

[0014] According to an embodiment of the present invention, referring to FIG. 1, there is shown a printing apparatus **10** configured to digitally print directly on the surfaces of a container **12** while the container **12** is in a continuous and constant motion. The printing apparatus **10** includes a plurality of print heads **14** for printing directly onto a curved surface **12a** of the container **12** and a plurality of container holders **16** in a continuous automated operation. Each of the plurality of container holders **16** is configured to hold or retain the container **12** and is capable of providing both rotational movement and linear movement, as will be described in greater detail below. In the present illustrated embodiment, the containers **12** are shown as oval in general cross-section; but printing on containers having other non-flat cross-sectional shape is also contemplated.

[0015] Referring again to FIG. 1, the printing apparatus **10** further includes a printing wheel or turret **18** for rotating the containers **12** disposed within the plurality of container holders **16** about a turret vertical axis A, as will be described below, an infeed starwheel **20** is operable for transporting the containers **12** to the turret **18**, and a discharge starwheel **22** is operable for transporting the containers **12** from the turret **18**.

[0016] The circular turret **18** is configured to rotate in a counter-clockwise turret direction B about the turret vertical axis A with a constant speed. The turret **18** includes the plurality of container holders **16** mounted thereon for rotating the containers **12** to be printed. Each of the plurality of container holders **16** is disposed on the turret **18** with an evenly spaced distance therebetween.

[0017] A conveyor belt **24** moves in a longitudinal direction C and feeds the containers **12** to be printed to the infeed starwheel **20**. The infeed starwheel **20** rotates in a clockwise infeed direction D, which is opposing to the rotational direction of the turret **18**. Each of the containers **12** is transported to and retained in each corresponding container holder **16** via the infeed starwheel **20**. Once the printing on the surface **12a** of the container **12** is completed, the printed container **12** is transported from the container holder **16** via the discharge starwheel **22**. The discharge starwheel **22** rotates in a clockwise discharge direction E, which is opposing to the rotational direction of the turret **18**. After exiting from the container holder **16**, the printed container **12** continuously moves on the conveyor belt **24** in the longitudinal direction C.

[0018] As noted above, the container **12** that may be used in connection with the present printing apparatus **10** has a shape with curved surfaces, e.g., oblong, oval, etc. Non-limiting examples of the container **12** to be printed include polyethylene terephthalate (PET) and high density polyethylene (HDPE), which may be made with monolayer or multilayer plastic materials.

[0019] Referring to FIGS. 1 and 2, the plurality of print heads **14** are arranged outside of the turret **18** and in close proximity to the turret **18** such that a nozzle dispensing path is substantially tangent to a moving direction (e.g., counter-clockwise turret direction B) of the curved surfaces **12a** of the container **12** to be printed. Each of the plurality of print heads **14** is an inkjet print head having one or more nozzles **26** for dispensing printing ink. The one or more nozzles **26** are positioned such that they are substantially parallel with a container vertical axis F. In the depicted embodiment, each of the plurality of print heads **14** has one nozzle. Additionally, it is contemplated that the print head **14** is capable of moving horizontally to allow the curved printing surfaces **12a** of the container **12** to be at a constant velocity during the printing process. As will be described in further detail with respect to the embodiment of FIGS. 4-6, this is also advantageous in printing on round containers.

[0020] Referring particularly to FIG. 2, each of the plurality of container holders **16** is configured to provide both rotational movement and linear movement during the printing process as each container holder **16** moves continuously along the counter-clockwise turret direction D, as will be described in greater detail below.

[0021] Each of the plurality of container holders **16** includes an upper slide **28** having an upper mandrel **30** mounted thereon, a lower slide **32** having a lower mandrel **34** mounted thereon. A rotary actuator **36** and a linear actuator **38** are employed to move the container **12** with respect to the print head **14**, as will be described in below. Non-limiting examples of the rotary actuator and linear actuator are server motor and stepper motor.

[0022] The container **12** is secured and retained within its corresponding container holder **16** by the upper mandrel **30** and the lower mandrel **34**, while moving along the counter-clockwise turret direction **D** and while being printed by the plurality of print heads **14**. Specifically, the upper mandrel **30** and lower mandrel **34** engage with a top end portion of the container **12** and a bottom end portion of the container **12**, respectively, such that each end portion of the container **12** is mounted onto the upper mandrel **30** and lower mandrel **34**. The upper mandrel **30** and lower mandrel **34** are located on the upper slide **28** and lower slide **30**, respectively, at a length from the plurality of the print heads **14** such that a consistent distance is maintained from the printing surfaces **12a** of the container **12** to the plurality of print heads **14** during the printing process, as the container **12** rotates about the container vertical axis **F**. The container vertical axis **F** is parallel to the turret vertical axis **A**.

[0023] In the depicted embodiment, the lower mandrel **34** of each of the plurality of container holders **12** is operably connected to the rotary actuator **36**. The rotary actuator **36** is implemented under the lower slide **32** of the container holder **12**, as shown in FIG. 2, and the rotary actuator **36** provides rotational movement about the container vertical axis **F** during the printing process. The rotational movement ensures that during the printing process, the printed location of the printing surface of the container **12** is maintained substantially perpendicular to the stationary print head **14** during the printing process. Alternately, the rotary actuator **36** may be operably connected to the upper mandrel **30** of each of the plurality of container holders **12**.

[0024] The rotary actuator **36** is configured such that it allows the container **12** to be rotated in either a counter-clockwise direction **G** or clockwise direction **H** with various desired degrees (e.g., up to 360 degrees). In the depicted embodiment, as shown in FIG. 3, the container **12** rotates about the container vertical axis **F** in the counter-clockwise direction **G**.

[0025] The linear actuator **38** is included in each of the plurality of container holders **16** for providing linear movement (toward and away from the print head **14**) of the container holder **16**. The linear actuator **38** is integrally connected to the lower slide **32** (which is connected to the upper slide **28** via a connecting bar **40**) and disposed between the lower slide **32** and a top surface of the turret **18**, as shown in FIG. 2. During the printing process, as the turret **18** continuously rotates at a constant speed and as the container **12** rotates, the linear actuator **38** allows the container **12** to move towards and away from the print head **14** along a line **I**. The linear movement combined with rotational movement allows a consistent distance to be retained from the printing surface of the container **12** to the print head **14**, as the container **12** rotates and moves continuously along the moving direction (e.g., counter-clockwise turret direction **D**).

[0026] Referring to FIGS. 3A-3E, the printing process on the printing surfaces of the container **12** will be described. In the depicted embodiment, each container holder **16** having the container **12** therewithin enters a printing area with one side of the printing surface initially contacting with the nozzle **26** of the printing head **14**, as shown in FIG. 3A. Due to the curved printing surfaces **12a** of the container **12**, the printing apparatus **10**, specifically the container holder **16**, is configured to rotate about the container vertical axis **F** as the printing on the container **12** progresses. As the container **12** is rotated about the container vertical axis **F** in the counter-clockwise direction **G** at an angular speed with the turret **18** being continuously moving in the counter-clockwise turret direction **D**, the nozzle **26** of the print head **14** dispenses the ink jet and prints directly on the curved printing surfaces **12a** of the container **12** until the entire printing surface is printed in a desired fashion, as shown in FIG. 3B-3E. While the container **12** is being rotated during the printing process, the linear actuator **38** of the container holder **16** moves the container **12** towards

and away from the nozzle **26** of the printing head **14** along the line I to maintain a consistent distance from the printing surfaces of the container **12** to the print head **14**, allowing the printed materials on the curved surfaces to be in a good print quality (e.g., the clearness of images or texts). [0027] The printing process is continuous until the entire desired printing surface **12a** of the container **12** is printed by the printing head **14** in one continuous motion. The print head **14** completes the printing process as the other side of the printing surface of the container **12** is reached, as shown in FIG. 3E. During the printing process, the print head **14** may remain stationary or move horizontally with respect to the turret **18** as the container **12** rotates about the container vertical axis F and moves linearly towards and away from the print head **14** to maintain a substantially constant distance between the print head **14** and the location of printing on the curved surface. The rotational movement and linear movement of the container **12** ensure that the location of printing on the curved surfaces **12a** of the container **12** remains generally perpendicular to the print head **14** as the container **12** rotates during the printing process.

[0028] Accordingly, the combination of the rotational movement of the turret **18**, the rotational movement of container **12**, and the linear movement of container **12** provides a printing technique that may be achieved rapidly and continuously, and vastly improves printing quality on curved printing surfaces of the container **12**.

[0029] From the foregoing, it will be appreciated that a printing apparatus according to the present invention provides a printing technique for directly printing on a container and improving printing quality on the curved printing surfaces of the container.

[0030] Referring now to FIGS. 4-6, a further embodiment of the present invention is shown which is particularly advantageous in printing on round bottles.

[0031] The present embodiment is substantially similar to the embodiment described above. For simplicity, with respect to the present embodiment, similar reference numerals will be used to denote similar elements.

[0032] The printing apparatus **110** of the present embodiment includes a plurality of print heads **114** for printing directly onto a curved surface **112a** of a container **112** held on a plurality of container holders **116** (FIG. 5) in a continuous automated operation. Each of the container holders **116** is configured to hold and retain a container **112** and is capable of providing both rotational movement and linear movement similar to that described above.

[0033] In the present illustrative embodiment, containers **112** are shown to be round containers in cross section. However, the present embodiment is not limited thereto.

[0034] As with the above embodiment, the present embodiment of the printing apparatus **110** includes a wheel or turret **118** for rotating the containers **112** around an axis A in the direction of arrow B. An infeed star wheel **120** is operable for transporting the containers **112** to the turret **118** and a discharge star wheel **122** is operable for transporting the containers **112** from the turret **118**. A conveyor belt **124** moves in a longitudinal direction C and feeds the containers **112** to be printed into the infeed star wheel **120**. The discharge star wheel **122** accepts the printed containers and after exiting from the container holder **116**, the printed containers **112** continuously move on a conveyor belt **124** in the longitudinal direction C. The operation of the conveyor belt, turret and star wheels are substantially similar to that described above with respect to the previous embodiment.

[0035] Referring additionally to FIG. 5 specifically, each of the plurality of container holders **116** is configured to provide both rotational movement and linear movement during the printing process.

[0036] The construction and movement of the container holders **116** is substantially similar to that described above with respect to FIG. 2. Each container holder **116** includes an upper slide **128** with an upper mandrel **130** mounted thereto, a lower slide **132** having a lower mandrel **134** mounted thereto.

[0037] A rotary actuator **136** and a linear actuator **138** are employed to move the container **112** with respect to the print head **114** in a manner similar to that described above with respect to FIG. 2 during the printing process. As the turret **118** continuously rotates preferably, but not necessarily, at

a constant speed and as container **112** rotates, the linear actuator **138** allows the container to move towards and away from the print head (arrow R in FIG. 4). The linear movement combined with the rotational movement allows a constant distance to be maintained between the printing surface of the container **112** and the print head **114** as the container rotates and moves continuously along the moving direction. While a rotary and linear actuator are described, the same movement may be accomplished via servo motor, stepper motor, cams, pneumatics, etc.

[0038] In addition, with respect to round bottles, it has been found that constant distance between the printing surface of container **112** and the print head may be further facilitated by arranging the print head **114** to be movable in a back and forth linear direction L as the container rotates about turret **118**.

[0039] Different from the embodiment shown in FIG. 2, the embodiment of FIG. 5 mounts the print head **114** for longitudinal movement generally along the path of the container holder **116** about turret **118** as indicated by double headed arrows L in FIGS. 4-6. In FIG. 5 the movement is into and out of the page.

[0040] Referring specifically now to FIG. 6, the printing process for printing on surfaces **112a** of round containers **112** may be described. While substantially similar to that described above with respect to FIGS. 3A-E, in this embodiment the printing process incorporates movement of the print head **114** along with movement of the container holder **116**.

[0041] Each container **112** supported on the container **116** holder (not shown) enters a printing area so that one point of the surface **112a** of container **112** is positioned adjacent nozzle **126** of printing head **114**. A center line C defines a neutral position for the print head along path L. As shown in FIG. 6, the print head is initially positioned at a location to the left a distance of $D_{sub.1}$ from the center line C. At this position, the container **112** on the container holder **116** is positioned fully extended toward the circumference of the turret **118** defined as $r_{sub.1}$ in FIG. 6. This defines position (I) where the nozzle **126** (not shown) is a distance P from container surface **112a**. As the container **112** moves about the turret and is rotated counter-clockwise by the container holder **116**, the print head **114** prints on the surface **112a** of the container. The container **112** also retracts away from the circumference of the turret from its fully extended position $r_{sub.1}$ to a position $r_{sub.2}$ (which is less than $r_{sub.1}$) defining position (II) where at that point the container **112** has been printed on 90° of the container surface **112a**. The print head **114** moves along path L with the movement of container so as to maintain the constant distance P between the print head **114** and the container **112**. When the container is retracted to $r_{sub.2}$ the print head has moved to the right to a position which is a distance $D_{sub.2}$ (less than $D_{sub.1}$) from the center line C.

[0042] As the container **112** continues to move about the turret **118**, it is retracted further to a position $r_{sub.3}$ which is less than $r_{sub.2}$ defined at position (III). Since the container **112** is continually rotating in a counter-clockwise direction, the container **112** will be printed on 180° of the container surface at position (III).

[0043] The print head will now be moved along path L to a distance $D_{sub.3}$ which is at the center line C so as to maintain the constant distance P.

[0044] The container **112** continues to move about the turret **118** from the position (III) to (IV) where, due to the counter-clockwise rotation, the container is now printed on 270° of the container. To maintain the constant distance P, the container is extended to $r_{sub.4}$ which is equal to $r_{sub.2}$. and the print head moves back to a position which is a distance $D_{sub.4}$ from center line C which is equal to distance $D_{sub.2}$ but in the opposite direction.

[0045] Still further, the container moves along the turret to a position (V) where the container is extended to a radius $r_{sub.5}$ which is equal to $r_{sub.1}$. As the container continues its counter-clockwise revolution it is now printed on 360° of the container. Similarly, the print head has now moved a distance $D_{sub.5}$ which is equal to distance $D_{sub.1}$, but in the opposite direction so as to maintain the constant distance P between the print head and the container D.

[0046] While the present embodiment shows printing 360° about the container **112**, it may be

appreciated that printing may take place on less than the entire circumference of the container.

[0047] Thus by movement of the container toward and away from the circumference of the turret while rotating the container and by movement of the print head a relatively constant distance is maintained between the print head and the container surface during printing.

[0048] In general, the foregoing description is provided for exemplary and illustrative purposes; the present invention is not necessarily limited thereto. Rather, those skilled in the art will appreciate that additional modifications, as well as adaptations for particular circumstances, will fall within the scope of the invention as herein shown and described and of the claims appended hereto.

Claims

1. (canceled)

2. An apparatus for printing on a plurality of containers, the apparatus comprising: a plurality of print heads configured to print directly on a curved surface of each of the plurality of containers; and a plurality of container holders for retaining the plurality of containers, each of the plurality of container holders configured to rotate a container retained therewithin and to move the container linearly relative to the plurality print heads in a continuous motion to maintain a print location on the curved surface of each of the plurality of containers at a substantially constant perpendicular distance from each of the plurality of print heads during a printing process; and wherein the plurality of print heads are configured to move horizontally in a back and forth moving direction generally along the path of the plurality of the containers.

3. An apparatus of claim 2, wherein the curved surfaces of the container are at a constant velocity during the printing process.

4. An apparatus of claim 2, each of the plurality of container holders includes a rotary actuator and a linear actuator for moving said container relative to said at least one print head.

5. An apparatus of claim 4, wherein said rotary actuator and said linear actuator provide for rotational movement of said container and movement of said container toward and away from said print head.

6. An apparatus of claim 5, wherein said rotational and linear movement of said container occurs along with said horizontal movement of the print heads.

7. An apparatus of claim 6, wherein said print heads maintain a substantial constant distance from the surface of said container.

8. An apparatus of claim 2, wherein said curved surface is cylindrical.

9. An apparatus for printing on a plurality of containers, the apparatus comprising: at least one print head configured to print directly on a curved surface of each of the plurality of containers; and a circular turret for rotating around a circumference thereof a plurality of container holders mounted thereon, the plurality of container holders for retaining the plurality of containers, each of the plurality of container holders configured to rotate a container retained therewithin and to concurrently move the container linearly towards and away from the at least one print head to maintain a predetermined distance from the at least one print head during a printing process; and wherein the at least one print head is configured to move linearly in a back and forth direction generally along an arcuate path of the plurality of the containers moving on the circular turret.

10. An apparatus of claim 9, further including an infeed starwheel for moving containers onto the circular turret and a discharge starwheel for moving containers off the circular turret.

11. An apparatus of claim 9, wherein the movement of the plurality of containers along the circumference of the turret and the linear movement of the at least one print head are configured so that during the printing process the container has no relative motion with respect to the at least one moving print head based on the concurrent movement of the container and at least one print head.

12. An apparatus of claim 9, wherein each of the plurality of container holders includes a rotary actuator and a linear actuator for moving said container relative to the at least one print head.

13. An apparatus of claim 9, wherein said curved surface is cylindrical.

14. An apparatus of claim 9, wherein the movement of the plurality of containers around the turret, movement of the containers toward and away from the circumference of the turret and the linear movement of said at least one print head tangential to the circumference of the turret maintains said predetermined distance between a container surface from said at least one print head during said printing process.
